Assignment 5

1.Create a table comparing at least three types of operating systems (such as Windows, Linux, and Android) based on their main features, target devices, and examples of apps you use on each.

Ans-

Operating System	Main Features	Target Devices	Examples of Apps
Windows	GUI, multitasking, wide hardware/software support, Windows Defender	Laptop, Desktop, PC	MS Word, Chrome, VLC
Linux	Open-source, secure, customizable, command-line support	Servers, PCs, laptops	Firefox, LibreOffice, VLC
Android	Touch interface, mobile apps, notifications, Google Play Store	Smartphones, tablets, smart TVs	Instagram, WhatsApp, Zomato
Your observation	Write what you noticed	Write device used	Write apps you use

2.Identify the operating system running on your phone and your laptop/desktop. Write down two unique features of each OS that affect your daily use of popular apps like Instagram or Zomato.

Ans-

1. Operating System on My Phone

Device: Samsung Galaxy A13
Operating System: Android

Two features that affect my daily use of apps like Instagram and Zomato:

1. **Google Play Store:** Android provides the Google Play Store, which allows me to easily install, update, and manage apps such as Instagram and Zomato.
2. **App Permissions:** Android allows me to control permissions such as **Location, Camera, Microphone, and Storage**. For example, Zomato can use my location to show nearby restaurants, while Instagram can use the camera and microphone for photos, videos, and reels.

2. Operating System on My Laptop/Desktop

Device: Laptop/Desktop
Operating System: Windows

Two features that affect my daily use of apps like Instagram and Zomato:

1. **Multitasking:** Windows allows me to run multiple applications and browser tabs at the same time. For example, I can use Instagram or Zomato in Chrome while working in MS Word or another application.
2. **Windows Security:** Windows includes built-in security features such as Microsoft Defender, which helps protect the laptop from malware and unsafe files while I use websites and applications.

3.List three security risks that are specific to mobile operating systems like Android or iOS, and explain how these risks could impact apps you use (for example, WhatsApp or Paytm).

Ans-

Security Risk	How the Attack/Risk Works	Impact on Apps
Malicious/Fake Apps	A user installs an application from an untrusted source. The app may contain malware.	Could steal information, SMS/OTP or account credentials used with apps such as Paytm.
Excessive App Permissions	An app gets unnecessary access to camera, microphone, contacts, location or files.	Private WhatsApp chats, contacts or other personal information could be exposed.
Phishing & Fake Login Pages	A fake message or website tricks the user into entering credentials or OTP-related information.	Attackers may attempt to take over accounts or perform fraudulent transactions.

4.Research and describe how a cloud-based operating system (such as Chrome OS or a virtual desktop environment) is different from a traditional desktop OS. Give one example of how this difference changes the way you access apps like YouTube or Flipkart.

Ans-

A cloud-based operating system, such as ChromeOS, is designed to use web applications and cloud services extensively. A traditional desktop operating system, such as Windows, is designed to run many applications directly on the computer.

Feature	**Traditional Desktop OS – Windows**	**Cloud-Based OS – ChromeOS**

Application usage	Many applications are installed and run locally on the computer.	Many applications are accessed through a web browser or cloud service.
Storage	Files and applications can be stored on the computer's SSD/HDD.	Greater use of cloud storage such as Google Drive.
Internet dependency	Many installed applications can work without an Internet connection.	Many services and applications work best when connected to the Internet.
Updates	Users may need to update applications and the operating system.	ChromeOS is designed to receive automatic system updates.
Example	Windows 11	ChromeOS

YouTube

On a **Windows laptop**, I can open YouTube in Chrome or another browser. I can also install and use many other applications directly on Windows.

On **ChromeOS**, I can open the Chrome browser and access **YouTube directly as a web-based service**. This means I do not need to install a traditional desktop YouTube application to watch videos.

Flipkart

I can open **Flipkart in the Chrome browser** on both Windows and ChromeOS. However, ChromeOS is more focused on accessing services through the web and cloud rather than installing traditional desktop applications.